

Evaluating protein binding interfaces with transformer networks

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Computational protein-binding studies are widely used to investigate fundamental biological processes and facilitate the development of modern drugs, vaccines and therapeutics. Scoring functions aim to assess and rank the binding strength of the predicted protein complex. However, accurate scoring of protein binding interfaces remains a challenge. Here we show that our evaluating Protein binding Interfaces with Transformer Networks (PIsToN) approach can distinguish native-like protein complexes from incorrect conformations. Protein interfaces are transformed into a collection of two-dimensional images (interface maps), each corresponding to a geometric or biochemical property. Pixel intensities represent the feature values. A neural network was adapted from a popular vision transformer with several enhancements: a hybrid component to accept empirical-based energy terms, a multi-attention module to highlight essential features and binding sites, and the use of contrastive learning for better ranking performance. The resulting PIsToN model substantially outperforms state-of-the-art scoring functions on well-known datasets.

With the advent of the machine learning (ML)-based tool AlphaFold¹, protein structure prediction has become more tractable. The next challenge in this domain is predicting protein complex structures². If the structures of individual target proteins are available, the complex can be predicted with molecular-docking tools. Given two protein structures, docking approaches aim to computationally predict the best binding locations and conformations forming stable complexes. Protein-docking tools are critical for the successful development of drugs, vaccines and therapeutics^{3,4}. Virtual screening saves financial and labour resources in drug design^{5,6}. However, modern computational tools for docking predict a large number of candidate complexes with seemingly good binding scores that fail confirmation in the laboratory⁷, suggesting inadequate scoring functions.

The accuracy of molecular-docking tools depends on the reliability of scoring functions. The first docking stage involves predicting many candidate conformations, either by fast Fourier transform correlations or by direct search⁸. During the second stage, each docking

conformation is ranked with a scoring function, with predictions favouring top-ranked conformations⁹. The best docking tools have the optimal binding conformation within the top-100 predicted docking structures with high frequency but less frequently in the top 10 and rarely in the top position¹⁰. Furthermore, docking programs routinely predict binding sites for molecules that do not interact in nature, leading to high false-positive rates¹¹. Therefore, improving scoring and evaluation functions would be tremendously useful.

Related work

Accurate evaluation of protein complexes is a challenging problem¹². Approaches with force fields attempt to evaluate complex atomic interactions¹³ but require approximations and result in reduced accuracies^{14,15}. 'Empirical-based' scoring functions include additional terms for better characterization of protein interactions, such as solvent-accessible surface area¹⁶ and counts of the rotatable bonds¹⁷, weighting them using linear regressions¹⁸. They are generally better than force-field methods

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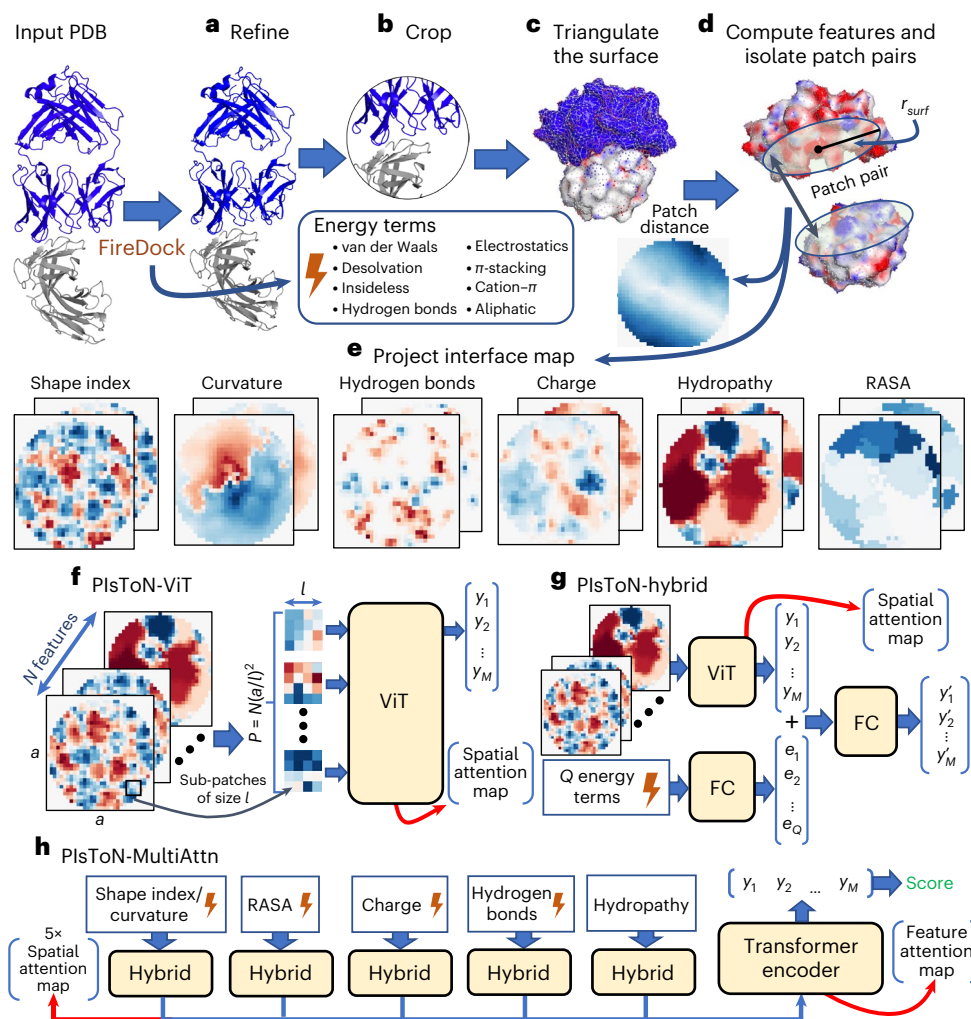


Fig. 1 | PlsToN framework. **a–e**, Feature engineering process for PlsToN (with an example on PDB ID 3B9K; ref. 85). **a**, Executing the FireDock to refine a PDB structure and compute empirical energy terms. **b**, Cropping protein complexes by the radius r_{surf} near the protein interaction centre. **c**, Triangulating the protein surface. **d**, Computing features associated with each point on a surface and isolating circular patches with the radius r_{surf} . **e**, Projecting patch features into a 2D interface map. **f–h**, Components of the PlsToN network to predict the binding class ('positive' or 'incorrect'). **f**, The ViT component that takes interface maps as input. **g**, The hybrid component, where representation from the ViT (top) is

combined with empirical energy terms (bottom), such that the network is aware of detailed surface representation and global approximated energies. FC, fully connected network. **h**, The final PlsToN network (multi-attention), where related interface maps and energy terms are combined into groups processed by independent hybrid networks. The output(s) are subsequently combined by the transformer encoder to produce the final score. The network provides a dual axis of explainability—spatial and feature attention points. The former is important surface regions, while the latter highlights the important feature groups.

but yield poor correlations with experimentally measured affinities and often fail to distinguish native interacting proteins from decoys¹⁹.

ML approaches can improve the scoring accuracy of protein binding. ML models learn the 'unknown' functions mapping protein complexes to their binding energies. Unlike empirical-based methods, ML methods learn nonlinear dependencies between energy terms, which is necessary for non-covalent forces²⁰. Also, ML methods avoid the reliance on our incomplete knowledge of force fields and prevent propagating errors from approximations²¹.

Recent proposals for representations for protein structures include using empirically computed energy terms as a direct input into ML models, such as support vector machine^{22,23} or random forest²⁴. Such models address nonlinear dependency issues, but their accuracy remains limited by approximations from empirically computed energy terms. A second set of methods includes features such as the frequency of each type of atom pair that come within some threshold distance of each other^{25,26}. Combining empirical-based

energy terms with contact frequency-based features improves the scoring accuracy²⁷ but is far from ideal, possibly because they lack three-dimensional (3D) complementary information from the proteins that affect binding stability²⁸.

Other studies incorporating detailed atomic spatial information isolate a 3D voxel grid around the interface region and apply convolutional neural networks (CNNs)^{29–33}. Examples include tools like DOCKing decoy selection with Voxel-based deep neural nEtwork (DOVE)³², DeepRank³³ and Deep Local Analysis (DLA)³⁴, but are prone to high false-positive rates⁹ and overfitting³⁵.

In graph-based approaches, a graph neural network (GNN) is used to learn binding embeddings. Vertices in the GNN correspond to chemical or geometric features (for example, atoms or amino acids), and edges correspond to contacts. The graphs can be constructed from molecular surfaces to reduce complex 3D information into compact signatures while preserving binding-related spatial features. Examples include iScore³⁶, PatchBag³⁷, GNN-DOVE³⁸ and DeepRank-GNN³⁹.

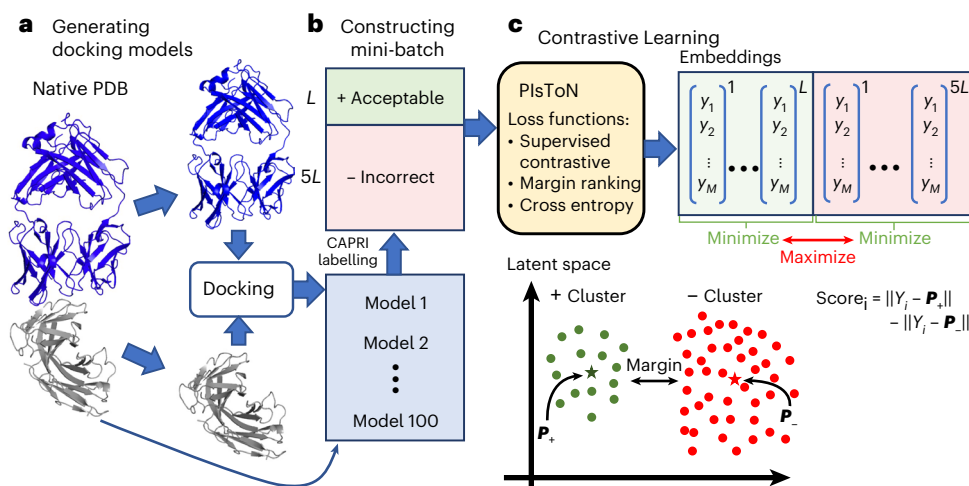


Fig. 2 | PIsToN training process. **a**, Generating 100 ‘acceptable’ or ‘incorrect’ docking models from a given native PDB structure from a training list. **b**, Constructing a mini-batch of ‘acceptable’ and ‘incorrect’ models with a 1:5

ratio. **c**, Update the weights of the PIsToN using contrastive loss functions to cluster embeddings of the same class near corresponding centroids while maximizing the distance between the clusters.

Molecular surface interaction fingerprint (MaSIF) adapted a ‘patch’ data representation to predict protein interactions⁴⁰. A patch was defined as a region on a solvent-excluded protein surface with a fixed geodesic radius around a potential contact point. Each point on the surface is associated with geometric and physico-chemical features. A Siamese graph convolutional network was trained to minimize the distance between embeddings from interacting patches while maximizing it for non-binding ones. MaSIF reportedly is 1,000-times faster than existing docking tools with only a slight loss of accuracy when compared with standard docking tools⁴⁰. The differentiable molecular surface interaction fingerprinting (dMaSIF) approach further enhanced the method with more efficient computation and sampling of the molecular surface⁴¹.

Shared limitations between MaSIF and dMaSIF, when addressed, could notably boost their performance efficacy. First, the training method generates negative patch pairs by randomly selecting surfaces outside the interface region. Therefore, negative instances often consisted of easily distinguishable non-complementary patches. We hypothesized that better training could be achieved using near-native non-interacting patch pairs. In addition, the network architecture consists of only convolutional layers, while better options, such as attention-based models⁴², can be considered.

Our contributions

Our proposed approach to evaluate Protein Interfaces with Transformer Network (PIsToN) has the following novel contributions.

Our approach uses a vision transformer (ViT) for protein binding assessment. This study pioneers the application of artificial visual intelligence (vision AI) tools for addressing the challenge of protein binding assessment. This has been achieved by modelling binding interfaces as two-dimensional (2D) multi-channel images and subsequently leveraging the vision transformer network⁴³. The surface patch representation was projected into images with pixel intensities corresponding to feature values associated at 1 Å resolution. PIsToN considers pairs of patches from binding interfaces, thus explicitly incorporating interaction properties, such as the distance between atoms, relative accessible surface area (RASA), van der Waals interactions, complementary surface charges, hydrophobicity values and more.

In our approach, ViT is enhanced with multiple attention axes, one for each feature type. The latent representations of each property were learned with an independent ViT network and combined in latent space using the transformer encoder.

Our approach uses physics-informed ML. ViT was augmented with a hybrid component to combine additional empirical energy terms. Consequently, beyond the image-based surface representation, the network also captures data related to approximated protein binding energies, providing a more comprehensive contribution from physics-based models.

Our approach allows for explainability. The multi-attention ViT allows for explainability in two ways: feature classes and binding sites essential for its decision-making.

Our approach uses contrastive mini-batch training. Contrastive learning with a novel loss function was introduced to learn discriminative embeddings for native binders and decoys. Instead of positive-random pairs⁴⁰ or a mixed batch of multiple positive and negative protein complexes^{33,36}, each training iteration consists of batched views of acceptable and incorrect binding poses of the same complex. The combination of supervised contrastive, margin ranking and binary cross-entropy terms in the loss function has resulted in superior embeddings.

Finally, our approach shows superior performance. As described later, the PIsToN model substantially outperforms state-of-the-art methods on three benchmark datasets.

Results

PIsToN framework

A novel and accurate framework to evaluate Protein Interfaces with Transformer Network (PIsToN) was developed (Fig. 1). The binding between two proteins is represented as an interface map, a projection of surface interface features into a multi-channel image. The enhanced hybrid multi-attention ViT was trained to distinguish between ‘acceptable’ and ‘incorrect’ binding classes.

The preparation of interface maps includes side-chain refinement, computing empirical-based energy terms, triangulating the surface, computing features on surface points, isolating a pair of circular patches and projecting them into a 2D image (Fig. 1a–e). The size of each interface map is fixed and defined by a geodesic radius r_{surf} (Fig. 1d). The surface features include shape index, curvature, hydrogen bonds, charge, hydrophobicity, shape index, RASA and distance between interacting surfaces (Fig. 1e).

The PIsToN multi-attention (PIsToN-MultiAttn) model groups related surface features and empirical energy terms into five branches: shape, RASA, charge, hydrogen bonds and hydrophobicity (Fig. 1h). Each

branch consists of an independent hybrid component that encodes interface maps using ViT and combines them with corresponding energy terms using a fully connected layer (Fig. 1f,g). The proposed network provides two axes of explainability—spatial attention highlights binding sites, and features attention suggests essential protein properties.

PlsToN was trained with contrastive learning for better ranking performance (Fig. 2). The training set consisted of 5,123 native protein complexes used by MaSIF-Search⁴⁰. Using the macromolecular docking tool HDock-light⁴⁴, we generated 17,291 with at least ‘acceptable’ quality and 452,031 ‘incorrect’ docking models⁴⁰ (Fig. 2a). The labels were assigned according to Critical Assessment of Predicted Interactions (CAPRI)⁴⁵. Each mini-batch consisted of multiple ‘acceptable’ and ‘incorrect’ docking models of the same protein complex (Fig. 2b). The weights were updated in accordance with the hybrid loss function that combined supervised contrastive⁴⁶, margin ranking⁴⁷ and binary cross-entropy losses. The method aims to cluster instances of the same class in embedding space while simultaneously pushing apart ‘acceptable’ and ‘incorrect’ docking models.

The details of PlsToN data preparation, architecture design, training and testing can be found in Methods.

Model selection

Each of the PlsToN components showed incremental improvement of the area under the receiver operating characteristic curve (AUC ROC) based on the MaSIF-test dataset (Table 1). The baseline AUC of the MaSIF architecture was 82.52% (Table 1, row 1). The CNN network trained with the concatenated 2D interface maps resulted in a competitive performance of AUC equal to 81.34% (Table 1, row 2). The CNN network had three blocks of convolutional and pooling layers, the sizes of which were constructed to match the MaSIF network (~66,000 trainable parameters). The ViT with spatial attention (Fig. 1f) improved the AUC to 83.14% (Table 1, row 3). Inclusion of the additional features corresponding to Euclidean distances between points on the two protein surfaces from a patch pair (patch distance) and RASA improved the AUC to 87.85% (Table 1, rows 4 and 5). The inclusion of empirical energy terms via the PlsToN-Hybrid component (Fig. 1g) increased the AUC to 90.36% (Table 1, row 6). The multi-attention architecture that matches energy functions with the corresponding physico-chemical category of features (Fig. 1h) boosted the AUC by ~2.5% (Table 1, row 7). The increase of the patch radius from 12 Å to 16 Å boosted the performance by another ~0.4% (Table 1, row 8). Finally, the implementation of the contrastive learning resulted in an AUC of 93.55% (Table 1, row 9). Further increase of the patch radius from 16 Å to 20 Å did not improve the AUC, possibly due to overfitting (Table 1, row 10). The model from row 9 had the best AUC ROC of 93.55 and was the chosen model for PlsToN.

In an effort to mitigate possible training bias attributed to the training/test split, we used BLAST⁴⁸ to filter out 107 (sequence identity ≥30%)⁴⁹ out of the 678 complexes for testing. The resulting change in AUC ROC (93.55% to 93.32%) was marginal, suggesting that the bias is negligible.

PlsToN scores

A PlsToN score for a given instance is computed as the difference in distances of the corresponding PlsToN embeddings of the positive and negative centroids (Fig. 2c). Therefore, lower values correspond to better binding. The maximum distance between two vectors in the latent space is two because the embeddings were normalized to a unit length (L2 normalization). Thus, the ideal binding corresponds to a score of -2 where the distance to a positive centroid is zero and the distance to a negative centroid is two. In contrast, a score of 2 corresponds to the most unlikely binding interface.

The PlsToN scores accurately discriminated between native complexes and incorrect docking models (Fig. 3a). The high-quality docking predictions had a distribution of scores similar to that for

Table 1 | Performance improvement of PlsToN models evaluated on the MaSIF-test dataset

Model	Improvement	r_{surf}	Parameters	AUC ROC (%)
(1) MaSIF-Search	None	12 Å	66,080	82.52
(2) PlsToN-CNN	2D interface maps	12 Å	68,822	81.34
(3) PlsToN-ViT	Spatial attention	12 Å	16,818	83.14
(4) PlsToN-ViT	Feature patch distance	12 Å	17,074	85.26
(5) PlsToN-ViT	Feature RASA	12 Å	17,586	87.85
(6) PlsToN-Hybrid	Energy terms	12 Å	18,248	90.36
(7) PlsToN-MultiAttn	Feature attention	12 Å	90,746	92.74
(8) PlsToN-MultiAttn	Patch radius change	16 Å	92,986	93.18
(9) PlsToN-MultiAttn	Contrastive learn	16 Å	92,986	93.55
(10) PlsToN-MultiAttn	Patch radius change	20 Å	95,864	93.32

native complexes (Fig. 3a, blue violins), while medium and acceptable categories had a larger overlap with incorrect models (Fig. 3a, purple and grey violins).

The default classification threshold value for PlsToN is zero (Fig. 3a, T1 dotted line) but can be adjusted depending on the dataset and the desired ratio between precision and sensitivity. To find the optimal classification boundary for testing datasets, we computed the Matthews correlation coefficient for different threshold values. The final classification threshold corresponded to the maximum Matthews correlation coefficient, averaged across ten splits, as suggested previously¹¹. The optimal thresholds were 0.85, -0.5 and -0.25 for MaSIF-test, CAPRI-score and PDB-2023, respectively (Fig. 3a, T2–T4 dotted lines). The same procedure for threshold optimization was performed for all other benchmark tools described in the next section (details in Supplementary Table 2).

PlsToN attention values and explainability

The attention maps and the range of attention scores obtained from the PlsToN network allow us to analyse the essential features and binding sites that led to a classification decision. When applied to the MaSIF-test set, each feature branch of the PlsToN multi-attention component had a different distribution of attention values (Fig. 3b). The network paid the greatest attention to geometric features (shape index and curvature) when predicting ‘positive’ complexes (Fig. 3b, top) and to RASA when predicting complexes as decoys or ‘incorrect’ (Fig. 3b, bottom).

All attention values had similar numerical ranges (16–26%), suggesting the essentiality of each protein property for binding. However, the property could be of interest if the corresponding attention is in the upper tail of the distribution (Fig. 3b). We considered a feature significant if the corresponding z-score of the attention value in the distribution is ≥1.96 (that is, $P \leq 0.05$). The same logic was applied to identify significant pixels from the spatial attention map. Figure 3b–d shows an example of how to use attention values and maps for achieving explainability. The example is the homodimeric complex of the protein phosphogluco isomerase (Protein Data Bank (PDB) ID 1J3R, chains A and B) from the archaeon *Thermococcus litoralis*⁵⁰. Our network correctly predicted the native structure of the complex as ‘positive’ with a score of -1.91. The attention value for hydrophobicity was identified as significant ($P = 0.04$), while the rest of the features were within the normal distribution (Fig. 3b, top row, turquoise stars). We further explored the interface map of interacting proteins corresponding to hydrophobicity and identified the region of significant pixels from the spatial attention (Fig. 3c, top row). After mapping the pixels from the interface map to the original structure, we found that significant pixels corresponded to residues 128 (isoleucine) and 101 (phenylalanine) of chain A and 150–152 (leucine and alanine) of chain B (Fig. 3d, top row). All of these residues are hydrophobic and located in

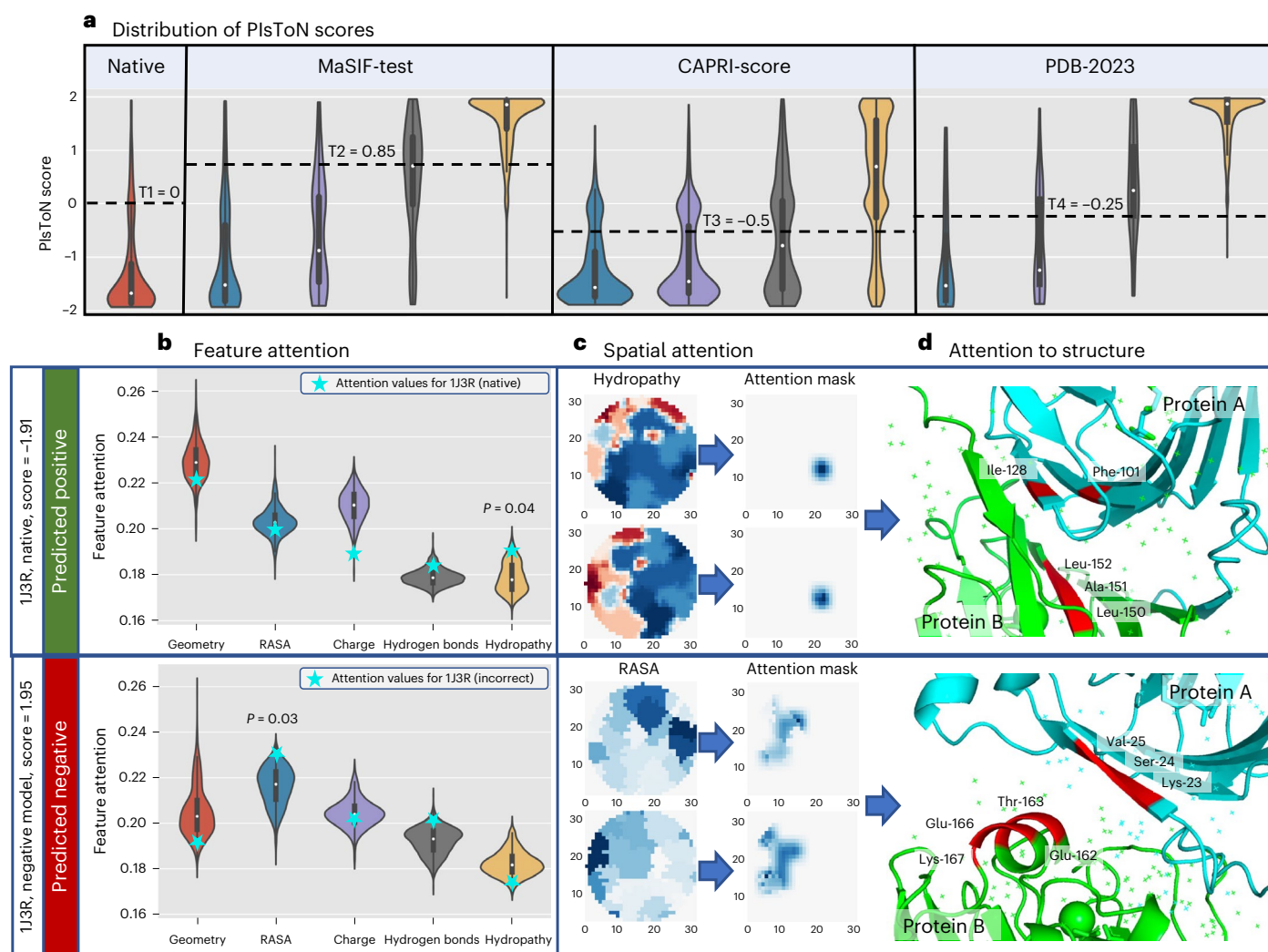


Fig. 3 | PlsToN explainability. **a**, Violin plots showing the distribution of PlsToN scores applied to native PDB complexes from the MaSIF-test set and docking models from the MaSIF-test, CAPRI-score and PDB-2023 sets. All complexes are subdivided into ‘native’ (red violin), ‘high’ (blue violin), ‘medium’ (purple violin), ‘acceptable’ (grey violin) and ‘incorrect’ (yellow violin) according to the CAPRI criteria. The dotted line shows our recommended thresholds: $T1 = 0$ if the dataset distribution is unknown, $T2 = 0.85$ if positive and negative instances are expected to be balanced, and between -0.5 and -0.25 ($T3$ and $T4$) if the data are expected to contain challenging examples with a high rate of false positives. **b**, Violin plots showing the distribution of PlsToN attention values for five interface map features of complexes classified as positive (top) and incorrect (bottom). Each violin is

colored according to the feature type: red for geometry, blue for RASA, purple for charge, grey for hydrogen bonds and yellow for hydrophobicity. In the violin plots in **a** and **b**, the white dots show the median value, the black boxes represent the first and third quartiles, and the vertical lines (whiskers) represent the $1.5 \times$ interquartile range. **c**, An example of identifying significant pixels from spatial attention map for PDB 1J3R complex. The color of each pixel signifies the value of the corresponding feature on a patch (blue for positive and red for negative, with darker colors signifying higher values). **d**, Structure of 1J3R at the interface, where significant residues are highlighted in red. The feature attention values (**b**, turquoise stars), spatial attention maps (**c**) and structure (**d**) are shown for the 1J3R native complex (top) and incorrect docked model (bottom).

the interaction region, suggesting a high role of the hydrophobic effect in the complex stability. Next, we analysed a ‘decoy’ docked model of the same protein pair (Fig. 3b–d, bottom row). The PlsToN network predicted a score of 1.95, suggesting improbable binding. The network paid the most attention to a RASA feature (Fig. 3b, bottom row). The spatial attention module identified the larger region of significance (Fig. 3c, bottom row), which was close to the contact point on a complex (Fig. 3d, bottom row). The likely explanation is that the highlighted residues are surface accessible on both proteins but too far away, possibly destabilizing any complex that may be possible at that site.

Comparison with benchmark tools

Our PlsToN model was competitive or better in all classification metrics (Table 2). The MaSIF-test contained 678 acceptable and incorrect

complexes and aimed to evaluate the classification performance (Table 2, MaSIF-test). The second dataset, called the CAPRI-score set, contained a large number of docking models for each pair of targets, therefore, evaluating the ability to correctly rank binding conformations (Table 2, CAPRI-score). The PlsToN had the highest AUC ROC and F1 scores on all testing datasets, which we consider as our primary metrics because it includes the balance between precision and recall (Table 2, columns 3 and 6).

Considering that GNN-DOVE³⁸ showed a slightly better average precision (AP) and dMaSIF⁴¹ performed the best in terms of balanced accuracy (BA) and recall on the CAPRI-score set (Table 2, CAPRI-score), we proceeded to compare these tools further using an additional manually curated dataset called PDB-2023 (Table 2, PDB-2023). The construction of this dataset ensured the inclusion of only new proteins

Table 2 | Classification performance of protein binding scoring functions on three datasets, MaSIF-test, CAPRI-score and PDB-2023

Dataset	Method	AUC ROC	AP	BA	F1	Precision	Recall
MaSIF-test	PlsToN	93.55	94.41	85.25	84.59	88.55	80.97
	dMaSIF	89.81	89.67	81.34	81.60	80.49	82.74
	MaSIF	82.52	83.96	74.93	72.51	80.25	66.13
	FireDock	89.95	90.54	81.83	81.58	82.81	80.38
	AP-PISA	89.85	90.30	81.98	81.28	84.63	78.18
	CP-PIE	88.11	89.28	81.03	79.94	84.87	75.55
	PyDock	86.11	87.17	77.66	77.62	77.79	77.45
	ZRANK2	83.56	87.37	77.89	74.36	88.48	64.13
	RosettaDock	75.28	82.77	76.06	69.81	94.50	55.34
	SIPPER	74.52	72.23	68.64	70.32	66.80	74.23
CAPRI-score	PlsToN	81.25	32.07	73.55	41.97	29.79	70.99
	dMaSIF	78.53	26.63	73.80	37.63	24.02	86.78
	iSCORE	73.51	26.71	69.84	36.58	24.74	70.12
	DeepRank-GNN	76.01	25.67	69.47	35.97	24.13	70.65
	DeepRank	72.22	23.27	70.57	38.47	26.97	67.07
	HADDOCK	70.34	28.67	66.20	37.87	31.88	46.63
	GNN-DOVE	70.73	35.37	63.06	38.05	51.49	30.17
	DOVE	56.39	22.47	55.01	19.21	42.67	12.40
	PlsToN	93.75	69.11	76.07	64.71	84.61	52.38
PDB-2023	dMaSIF	91.86	35.02	67.49	36.65	36.80	36.51
	GNN-DOVE	53.02	5.14	52.76	6.53	4.62	11.11

The numbers in bold within each column denote the maximum value observed for the respective performance metric.

released after January 2023, thereby minimizing bias towards the training set. The structural alignment analysis indicated that PDB-2023 showed the least structural bias to the training complexes compared with MaSIF-test and CAPRI-score (Supplementary Fig. 3a). Most of the template modeling (TM) scores of structural alignment between pairs of complexes in PDB-2023 were below 0.4 (Supplementary Fig. 3b) and did not show any prominent clusters (Supplementary Fig. 6), indicating a high level of diversity. When applied to the PDB-2023, PlsToN had superior performance in all classification metrics, proving the ability of the model to generalize well on previously unseen pairs of proteins (Table 2, PDB-2023).

An analysis of success rate showed a better performance of PlsToN in the task of ranking the docking poses among many generated models of CAPRI-score and PDB-2023 datasets (Table 3). The CAPRI-score contained 16,581 binding poses generated from a tool HADDOCK (High Ambiguity Driven protein-protein DOCKing)⁵¹ from 13 CAPRI targets, where at least two docked models are correct³³. The PlsToN model was competitive with iScore³⁶ in identifying the correct docking pose in CAPRI-score as the top-1 prediction (Table 3, CAPRI-score). Starting from the top 10, our model outperformed every other benchmarked scoring function (Table 3, CAPRI-score, columns 4–7). Similarly, PlsToN outperformed dMaSIF⁴¹ and GNN-DOVE³⁸ on the PDB-2023 set, which consisted of 53 native protein complexes and 5,300 generated docking models (Table 3, PDB-2023).

Runtime analysis

The PlsToN preprocessing was more computationally efficient than the predecessor MaSIF⁴⁰ (Supplementary Fig. 1a). PlsToN computed patches with a radius 12 Å almost five times faster (Supplementary Fig. 1a, red and grey lines). Furthermore, PlsToN computed larger patches with the size of 16 Å and 20 Å faster than MaSIF's 12 Å patches

(Supplementary Fig. 1a, blue and purple lines). Such efficiency was achieved by removing (cropping) residues outside the interaction region that do not contribute to the properties of the computed patch. Similarly, PlsToN had a much faster inference time, possibly due to the memory efficiency of the PyTorch library (Supplementary Fig. 1b). While dMaSIF had the slowest inference time, the method had better overall runtime because it did not involve any feature preprocessing. All runtime tests were performed on a single central processing unit and 256 GB volume of random-access memory.

Discussion

The developed PlsToN deep learning model outperforms the current state-of-the-art empirical and ML-based protein scoring functions in identifying viable protein binding interfaces (Table 2).

Our method introduces several novel elements. First, we represent molecular surface interfaces as ML-friendly, multi-channel images (Fig. 1a–e). Our representation captures neighbourhood information in a manner similar to the geometric patch objects used in MaSIF, but allowed us to use best practices of vision AI. In addition, the set of features was extended to include the distance between interacting surfaces, RASA and empirical energy terms. Next, we introduce a novel hybrid multi-attention transformer network adapted from the popular image classification ViT network (Fig. 1f–h)⁴³. The hybrid component accepts empirically computed energy terms along with 2D interface maps. We adapted the ViT to include dual attention: spatial attention to highlight essential binding sites and feature attention to identify the relative importance of specific protein properties (Fig. 3). Finally, we introduce a novel contrastive learning training strategy for protein interface classification. For each native protein structure, the network learns prototype centroids of acceptable and incorrect binding pose representations to achieve maximum

Table 3 | Success rate performance

Dataset	Method	Top 1	Top 10	Top 25	Top 100	Top 200
CAPRI-score	PlsToN	38	69	76	76	100
	dMaSIF	15	38	53	76	92
	HADDOCK	23	46	53	69	76
	DeepRank-GNN	23	46	53	69	84
	DeepRank	15	53	61	69	92
	iSCORE	38	53	61	76	84
	GNN-DOVE	15	61	69	76	100
	DOVE	0	15	46	46	69
PDB-2023	PlsToN	86	94	98	100	100
	dMaSIF	50	84	94	100	100
	GNN-DOVE	3	18	33	100	100

The percentage of CAPRI targets for which at least one acceptable model is identified among top-1, top-10, top-25, top-100 and top-200 docking models. For each column, the numbers presented in bold signify the maximum value within each respective 'top' category.

separability between the classes (Fig. 2). Each innovative component showed incremental improvement in the model performance (Table 1).

PlsToN outperforms other protein interface evaluation methods in classification and ranking tasks. However, the high rate of false positives for every method suggests the need for further improvements. We showed that the success rate for the top-1 prediction of scoring CAPRI models is as low as 38% for PlsToN and the other best competitive tool, iScore³⁶ (Table 3, column 2). One possible reason for a limited ranking performance is the competition for binding sites. For example, if two proteins have several favourable binding spots, current approaches will fail to distinguish the one with the minimum energy (Supplementary Fig. 4). The PlsToN model predicts the viability of the protein binding interface and does not necessarily reflect the strength of the binding. The improvement can be achieved by additional training on the set of protein complexes with experimental binding affinities, such as PDB-bind⁵². Yet, when the top-10 predictions were considered, the PlsToN identified 69% of correct complexes in the CAPRI-score set and 94% in the PDB-2023 set, respectively, which is notably better than the competitors (Table 3, column 4).

The fact that PlsToN can rank the native binding complexes higher than other approaches suggests that our model will be of value in many docking applications, such as vaccine design and drug design. Given docking models for two protein targets, PlsToN has a higher chance of placing the correct configuration in the top-10 predictions, thus potentially speeding up the screening of protein–protein interaction assays.

The PlsToN can be extended to score protein binding complexes generated by tools for multimeric interface prediction from sequences, such as AlphaFold-Multimer⁵³. Another application can be molecular mimicry search, where a large set of antibody–antigen structures are scanned for cross-reactivity^{54,55}.

PlsToN's computational efficiency can save computational resources when thousands of candidates need to be screened (Supplementary Fig. 1). The patch computation time was substantially reduced compared with the predecessor MaSIF⁴⁰. Even though dMaSIF completely avoided the computation of physico-chemical features and improved the runtime substantially⁴¹, PlsToN showed superior performance, suggesting that the image-based approach has intrinsic value. The surface features, such as charge, hydrophobicity and electrostatics, explicitly encode physics into the input image and are valuable for ML as well as its explainability. As the patch computation for each protein complex is an independent process, the pre-computation is easily

parallelizable and can be accelerated with the use of a supercomputer for large-scale analysis.

The strong performance of PlsToN compared with other interface scoring methods suggests the importance of enhancing feature engineering and using architectures that reflect our understanding of the domain.

Methods

Interface maps

Given a protein complex structure in PDB format⁵⁶, we generate an interface map by projecting surface features from the binding interface of the two proteins to create a multi-channel image (Fig. 1a–e), where each channel is dedicated to a feature type. Transforming data into 2D images (that is, feature maps) before applying the ML model was successfully used for other ML applications by ref. 57. Such transformations work well because ML techniques like convolutions work best on images.

Towards this goal, the protein structures were first refined to incorporate side-chain flexibility (Fig. 1a). The restricted interface side-chain optimization with 50 Monte Carlo cycles was performed with FireDock (Fast Interaction Refinement in molecular Docking)⁵⁸. The binding free energy terms of the refined structures computed by FireDock were set aside as an additional input to the deep learning model (Fig. 1b). The following energy terms were considered: van der Waals⁵⁹, desolvation⁶⁰, insideness⁵⁸, hydrogen and disulfide bonds⁶¹, electrostatics⁶², and π -stacking, cation– π and aliphatic interactions⁶³.

Second, the protein complexes were cropped to within distance r_{surf} from the 'protein interaction centre' (Fig. 1b), which was computed as the geometric centre of contact points. Contact points were defined as the coordinates of atoms within 5 Å proximity of an atom of the binding protein. Cropping improved memory efficiency by discarding residues with negligible influence on binding.

Third, the solvent-excluded surface was triangulated and re-scaled to a granularity of 1 Å using the MaSIF data preparation module⁴⁰ (Fig. 1c).

Next, we computed features associated with a given pair of circular 'patches' on the interacting surfaces (Fig. 1d). A patch is defined as the set of vertices on a triangulated protein surface within a geodesic distance of r_{surf} from the interaction centre. We computed shape index, curvature, hydrogen-bond potential, charge and hydrophobicity features for each surface point on a surface using the MaSIF data preparation module⁴⁰. Next, we computed another image for patch distance, which is generated from a grid of Euclidean distances between points on the two protein surfaces that project corresponding points on the patch pair. Lastly, the RASA⁶⁴ was computed for each patch residue using DSSP v2.3⁶⁵.

Finally, patch features were converted into an image (interface map) with pixel intensities proportional to the feature values (Fig. 1e). Surface points on patches were projected onto a 2D plane using a multidimensional scaling algorithm⁶⁶, which reduces the dimensionality to two while approximately preserving geodesic distances between two points on the patch. The final surface image of a given feature was computed as a 2D grid of size $2r_{\text{surf}} \times 2r_{\text{surf}}$. The intensity of each pixel for a given feature f is equal to smoothed value over the four nearest neighbours:

$$f_{ij} = \frac{\sum_{k=1}^4 f_k d_k^{-1}}{\sum_{k=1}^4 d_k^{-1}},$$

where $i, j \in \{0, 1, \dots, 2r_{\text{surf}}\}$ are the coordinates on the projected 2D grid, f_{ij} is the intensity of the pixel (i, j) , k is the number of neighbouring points on the protein surface close to (i, j) , d_k is the Euclidean distance from (i, j) to the k th surface point, and f_k is the value of feature f of the k th surface point.

Network architecture

Figure 1f–h shows the ViT, hybrid and multi-attention components of the PlsToN network. PlsToN-ViT was adapted from the ViT architecture developed for image classification⁴³. The input to the model was interface maps of size a projected from protein surfaces (Fig. 1e). Instead of conventional RGB, our ViT network accepts N channels corresponding to the number of surface features (Fig. 1f). The image gets subdivided into squares of size l , resulting in $P = (a/l)^2$ mini-patches. Then, the set of squares gets linearly projected into a set of M -dimensional token embeddings. Similar to the original ViT, we appended a learnable class token to the set of embeddings. The output of the transformer encoder is an embedded vector $Y = \{y_1, y_2, \dots, y_M\}$ and spatial attention map that highlights mini-patches essential for the classification decision.

The PlsToN-Hybrid component was developed to combine empirically computed energies with the interface maps (Fig. 1g). The Q energy terms were projected onto a latent vector $\{e_1, e_2, \dots, e_Q\}$ using a fully connected network, which was then concatenated to the vector Y obtained from ViT. The merged vector was projected to a new vector Y' of size M , thus mapping the interface map and energy terms on to a latent space.

PlsToN-MultiAttn efficiently combines the ViT and hybrid components to include spatial and attention features. As shown in Fig. 1h, the input is subdivided into five branches of related interface maps and energy terms. Each branch contains an independent transformer network that learns separate latent vectors with spatial attention maps. The latent vectors of each branch are then aggregated with a class token into the transformer encoder for a final prediction. Also important is that the feature attention map is extracted to highlight the importance of each group. The first of the five feature groups is shape complementarity combined with van der Waals (attractive and repulsive) and an ‘insideness’ measure. Second, RASA was grouped with desolvation. Third, the charge feature was concatenated with electrostatic energies (attractive, repulsive, long range and short range), and π -stacking, cation– π and aliphatic interactions. Fourth, hydrogen-bond interaction maps were combined with hydrogen and disulfide bonding energies. The last one was the hydrophathy feature. As the interatomic distances affect the energies from each group, the patch distance feature was appended as an extra channel to each branch. The class probability was computed as the Euclidean distance to the class centroid (see ‘Training process’ for details).

Training process

For the training process, each native protein complex from the training set was separated and re-docked with HDock-light v. 1.1 to generate 100 docked models (Fig. 2a)⁴⁴. Each docked model was then compared with the native PDB structure to assess its quality. A docking model was labelled ‘acceptable’ if it satisfied the following CAPRI condition⁴⁵:

$$\text{Acceptable} = \text{FNAT} \geq 0.1 \text{ and } (\text{fRMSD} \leq 10 \text{ \AA} \text{ or } \text{iRMSD} \leq 4 \text{ \AA}),$$

where FNAT is a fraction of native contacts recovered, iRMSD is the root mean square deviation (RMSD) between the interface C_α atoms of the docked model and the native structure, and fRMSD is the global RMSD between the C_α atoms of the two structures. Models for which the condition failed were labelled as ‘incorrect’. We note that the way incorrect/negative examples are chosen is an important contributor to why our model performs better than MaSIF, which uses random patches to select the negative examples for training.

The training was performed in mini-batches, each consisting of multiple views of sets of acceptable and incorrect binding poses of a single protein complex (Fig. 2b). As HDock produces many more incorrect docking models than acceptable ones, a mini-batch is limited to a 1:5 ratio of acceptable to incorrect models to limit the imbalance. Each epoch was trained on patches from the same L acceptable and $5L$ randomly selected incorrect docking poses. At each training epoch, patch pairs were augmented with random rotations from 0° to 360° .

This form of data augmentation not only amplified the count of training instances but also compelled the network to maintain consistency in its score generation across rotations. The values of each pixel were normalized with standard scaling. The training was guided with a contrastive learning process (Fig. 2c). The loss function consisted of three terms—supervised contrastive⁴⁶, margin ranking⁴⁷ and binary cross-entropy. The supervised contrastive loss was first introduced for image representational learning. The method aims to cluster instances of the same class in embedding space while simultaneously pushing apart clusters of samples from different classes⁴⁶. The idea is similar to Siamese training but generalized on an arbitrary number of positive and negative examples in a batch. In our formulations, the supervised contrastive loss looks as follows:

$$L_{\text{supCon}} = - \sum_{i=1}^L \frac{1}{L} \sum_{j=1}^L \log \frac{\exp(Y_i + Y_j + \tau)}{\sum_{a \notin \{i\}} \exp(Y_i + Y_a + \tau)} - \sum_{i=1}^{5L} \frac{1}{5L} \sum_{j=1}^{5L} \log \frac{\exp(Y_i - Y_j - \tau)}{\sum_{a \notin \{i\}} \exp(Y_i - Y_a - \tau)},$$

where L is the number of acceptable docking models, Y_i is the embedding for the i th model of acceptable (+) or incorrect (−) protein complex, a is the set of instances excluding i th complex, and τ is the temperature hyperparameter that controls the clustering tightness.

To further coordinate separability, we introduced prototypical layers that serve cluster centroids (Fig. 2c). The coordinates of positive and negative prototypes (P^+ and P^-) were learned with margin ranking loss:

$$L_{\text{MR}} = \sum_{i=1}^{6L} \begin{cases} \max(0, \|Y_i - P_+\| - \|Y_i - P_-\| + \text{margin}), & i \in \{+\} \\ \max(0, -\|Y_i - P_+\| + \|Y_i - P_-\| + \text{margin}), & i \in \{-\} \end{cases}$$

The interface score was predicted as the difference in distances to positive and negative prototypes so that smaller values correspond to a better binding:

$$\text{Score}_i = \|Y_i - P_+\| - \|Y_i - P_-\|$$

The final loss was the linear combination of contrastive, margin ranking, and binary cross-entropy (BCE):

$$\text{LOSS} = L_{\text{supCon}} + L_{\text{MR}} + L_{\text{BCE}}$$

As the loss function involves computing distance in high-dimensional feature space, we performed the L2 normalization of embedding vectors to transform the latent space into the unit hypersphere⁶⁷.

The networks were trained with a maximum of 200 epochs with the early stopping condition of validation AUC not improving 10 consecutive times. The AdamW optimizer with the learning rate of 0.0001 and weight decay of 0.001 was used for weights optimization⁶⁸. The temperature τ that controls clustering tightness of contrastive loss was set to 0.5 (ref. 46).

Datasets

We tested PlsToN with three representative datasets as described below.

MaSIF dataset. The first dataset included the complexes of protein–protein interactions used to train and test MaSIF software⁴⁰. A minor fraction of complexes were excluded due to the memory overhead during the preprocessing step. In addition, 20% of the testing set was randomly subsampled into a validation set, which was used for the early stopping of PlsToN models. As a result, we preprocessed 4,942 training, 181 validation and 678 testing native protein complexes. In total, we generated 17,291 positive and 452,031 negative training complexes.

For validation, we obtained 703 positive and 16,864 negative docked structures. For testing, only one positive and one negative complex were randomly selected during the benchmark with other scoring functions, resulting in 678 positive and negative complexes.

CAPRI set. The 13 protein pairs from the CAPRI challenge set were used to test the ranking performance⁶⁹. The protein-docking models were extracted from the DeepRank study³³. The dataset included 16,581 complexes generated by the HADDOCK docking tool with corresponding CAPRI quality labels⁵¹, which were then used for testing.

PDB-2023 set. As a final test set, we chose protein complexes deposited in the year 2023 to the Protein Data Bank (PDB) maintained by the Research Collaboratory for Structural Bioinformatics (RCSB) organization⁵⁶. The 2023 complexes were processed with the intent of minimizing the bias from the training set of all the tools, which could not have included any recently determined complexes. The collection of protein complexes released in 2023 was obtained and manually curated from the RCSB PDB portal⁵⁶. Repetitive complexes and protein–ligand interactions were excluded from the set, resulting in 53 non-redundant heterodimers. The protein chains from each native complex were separated and re-docked using HDock⁴⁴, which resulted in 126 acceptable and 5,174 incorrect docking models used for testing.

The protein complexes from the three test datasets were evaluated to estimate the extent of bias in the performance caused by overlap with the training set. For each target protein complex, the maximum template modeling (TM)-score between that complex and any complex in the training set was computed (using US-align version 20220924⁷⁰) and used as a measure of ‘similarity’ to the training set (Supplementary Fig. 3a).

Benchmarking

Empirical-based tools. The empirical-based scoring functions can be subdivided into energy-based and potential-based methods. The binding score of the first category is defined as the weighted sum of energy terms. The following energy-based methods were considered: FireDock³⁸, PyDock⁷¹, Rosetta⁷² and ZRANK2⁷³. Each scoring function includes electrostatics, desolvation and van der Waals energy terms. FireDock and Rosetta include additional hydrogen and disulfide bonds. The unique term of FireDock is the internal energy that consists of bond stretching, angle bends and torsional twists. The RosettaDock includes additional side-chain rotamer energies. The potential-based methods involve computing the atomic and residue-level interaction properties, such as frequency of interaction types and solvent-accessible surface area. The following potential-based methods were considered: AP-PISA⁷⁴, SIPPER⁷⁵ and CP-PIE⁷⁶. All empirical-based scoring functions we executed on CCharPPI online server⁷⁷.

MaSIF-Search. Our work was compared with a 2D CNN-based MaSIF-Search. The pre-trained MaSIF-Search ‘sc05’ model was used with recommended parameters for data preprocessing⁴⁰. Circular patches of 12 Å radius were computed from the surfaces of interacting proteins using the MaSIF data preparation module. The data structure of the patch is a grid of 80 bins with 5 angular and 16 radial coordinates. Five features were extracted from each grid cell of a patch—shape index, distance-dependent curvature, electrostatics, hydrophathy and electrostatic potentials—resulting in an 80-dimensional descriptor vector. The network was trained such that the confidence of interaction between two patches is inversely proportional to the Euclidean distance between corresponding descriptors. The best minimum distance among the three most complementary patches in the complex was used as the measure for a binding score.

dMaSIF-Search. The pre-trained model ‘dMaSIF_search_3layer_12A_16dim’ was used for our benchmarks⁴¹. The dMaSIF extended MaSIF by bypassing the pre-computation of physico-chemical features and

instead calculating molecular surfaces directly from the atomic point cloud in real time. The input to the model has a similar data structure to a 12 Å patch, where each point on a surface is represented as a one-hot encoding of surface chemicals combined with Gaussian and mean curvatures. For each protein complex, we used the default dMaSIF threshold to determine whether two patches were in contact. Given that a protein complex typically consists of multiple contact points, we evaluated three distinct approaches for scoring the complex: calculating the average, minimum and maximum scores across all potential contacts. As a result, the average approach demonstrated the highest AUC on the MaSIF-test dataset (Supplementary Fig. 5). Therefore, the average of dMaSIF scores across contact points was utilized to evaluate binding conformations from all testing sets.

Other deep learning-based tools. The performance on the CAPRI set was compared with three other deep learning-based methods: iScore³⁶, DOVE³², DeepRank³³, GNN-DOVE³⁸, DeepRank-GNN³⁹ and dMaSIF⁴¹. The docking scores for the CAPRI-score set were extracted from the DeepRank studies^{33,39}.

Evaluation metrics. The predictive power was estimated with the AUC ROC and the area under the precision–recall curve. The docking models with high, medium and acceptable CAPRI quality were called ‘positive’, while the rest of the docking models were labelled ‘negative’. The ability to rank was evaluated with success rate, which is defined as the percentage of complexes for which at least one model of acceptable quality is found in the top-*N* selected models. Other classification metrics included balanced accuracy, F1 score, precision and recall. The classification threshold was chosen as the average of thresholds across ten splits with the maximum Matthews correlation coefficient, as proposed previously for the virtual screening application¹¹.

Software and hardware

The distance between atom coordinates was computed with KDTree from SciPy package v.1.1.0 (ref. 78). The FNAT, iRMSD and tRMSD were computed using pdb2sql Python v.0.5.2 (ref. 79). The neural networks were implemented in PyTorch v.1.10 (ref. 80). The interface maps were visualized with plotly v.5.10.0 (ref. 81). The pairwise sequence alignment was performed using Nucleotide–Nucleotide BLAST 2.2.28 (ref. 82). The protein structures were visualized in PyMOL 2.5.4 (ref. 83). The training and testing of deep learning models were performed on a GeForce GTX 1080 Ti 8 graphics processing unit, a 256 GB RAM, 28 core Intel Xeon central processing unit E5-2650, and also on Instructional and Research Computing Center resources at Florida International University (<http://ircc.fiu.edu>).

Ethics and inclusion statement

The roles and responsibilities were agreed on among the authors ahead of the research, and all capacity-building plans were discussed.

Data availability

The lists of training/testing protein complexes and pre-computed interface maps are available on Zenodo at <https://doi.org/10.5281/zenodo.7948337>.

Code availability

The PIsToN software and benchmark datasets are available under license from <https://biorg.cis.fiu.edu/piston/>. The specific version of PIsToN used in this study is readily available for access via Zenodo⁸⁴.

References

1. Jumper, J. et al. Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583–589 (2021).
2. Callaway, E. After AlphaFold: protein-folding contest seeks next big breakthrough. *Nature* **604**, 234–238 (2022).

3. Vakser, I. A. Protein–protein docking: from interaction to interactome. *Biophys. J.* **107**, 1785–1793 (2014).
4. Shin, W.-H., Christoffer, C. W. & Kihara, D. In silico structure-based approaches to discover protein–protein interaction-targeting drugs. *Methods* **131**, 22–32 (2017).
5. Meng, X.-Y., Zhang, H.-X., Mezei, M. & Cui, M. Molecular docking: a powerful approach for structure-based drug discovery. *Curr. Comput. Aided Drug Des.* **7**, 146–157 (2011).
6. Scior, T. et al. Recognizing pitfalls in virtual screening: a critical review. *J. Chem. Inf. Model.* **52**, 867–881 (2012).
7. Gupta, M., Sharma, R. & Kumar, A. Docking techniques in pharmacology: how much promising? *Comput. Biol. Chem.* **76**, 210–217 (2018).
8. Huang, S.-Y. Search strategies and evaluation in protein–protein docking: principles, advances and challenges. *Drug Discov. Today* **19**, 1081–1096 (2014).
9. Shen, C. et al. From machine learning to deep learning: advances in scoring functions for protein–ligand docking. *Wiley Interdiscip. Rev. Comput. Mol. Sci.* **10**, e1429 (2020).
10. Moal, I. H., Torchala, M., Bates, P. A. & Fernández-Recio, J. The scoring of poses in protein–protein docking: current capabilities and future directions. *BMC Bioinformatics* **14**, 286 (2013).
11. Adeshina, Y. O., Deeds, E. J. & Karanicolas, J. Machine learning classification can reduce false positives in structure-based virtual screening. *Proc. Natl Acad. Sci. USA* **117**, 18477–18488 (2020).
12. Li, J., Fu, A. & Zhang, L. An overview of scoring functions used for protein–ligand interactions in molecular docking. *Interdiscip. Sci. Comput. Life Sci.* **11**, 320–328 (2019).
13. Kortemme, T. & Baker, D. A simple physical model for binding energy hot spots in protein–protein complexes. *Proc. Natl Acad. Sci. USA* **99**, 14116–14121 (2002).
14. Liu, X., Peng, L. & Zhang, J. Z. Accurate and efficient calculation of protein–protein binding free energy–interaction entropy with residue type-specific dielectric constants. *J. Chem. Inf. Model.* **59**, 272–281 (2018).
15. Huang, S.-Y., Grinter, S. Z. & Zou, X. Scoring functions and their evaluation methods for protein–ligand docking: recent advances and future directions. *Phys. Chem. Chem. Phys.* **12**, 12899–12908 (2010).
16. Durham, E., Dorr, B., Woetzel, N., Staritzbichler, R. & Meiler, J. Solvent accessible surface area approximations for rapid and accurate protein structure prediction. *J. Mol. Model.* **15**, 1093–1108 (2009).
17. Eldridge, M. D., Murray, C. W., Auton, T. R., Paolini, G. V. & Mee, R. P. Empirical scoring functions: I. The development of a fast empirical scoring function to estimate the binding affinity of ligands in receptor complexes. *J. Comput. Aided Mol. Des.* **11**, 425–445 (1997).
18. Trott, O. & Olson, A. J. AutoDock Vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *J. Comput. Chem.* **31**, 455–461 (2010).
19. Chen, Y.-C. Beware of docking! *Trends Pharmacol. Sci.* **36**, 78–95 (2015).
20. Muley, L. et al. Enhancement of hydrophobic interactions and hydrogen bond strength by cooperativity: synthesis, modeling, and molecular dynamics simulations of a congeneric series of thrombin inhibitors. *J. Med. Chem.* **53**, 2126–2135 (2010).
21. Liu, J. & Wang, R. Classification of current scoring functions. *J. Chem. Inf. Model.* **55**, 475–482 (2015).
22. Kinnings, S. L. et al. A machine learning-based method to improve docking scoring functions and its application to drug repurposing. *J. Chem. Inf. Model.* **51**, 408–419 (2011).
23. Das, S. & Chakrabarti, S. Classification and prediction of protein–protein interaction interface using machine learning algorithm. *Sci. Rep.* **11**, 1761 (2021).
24. Zilian, D. & Sotriffer, C. A. SFCscore(RF): a random forest-based scoring function for improved affinity prediction of protein–ligand complexes. *J. Chem. Inf. Model.* **53**, 1923–1933 (2013).
25. Durrant, J. D. & McCammon, J. A. NNScore: a neural-network-based scoring function for the characterization of protein–ligand complexes. *J. Chem. Inf. Model.* **50**, 1865–1871 (2010).
26. Ballester, P. J. & Mitchell, J. B. A machine learning approach to predicting protein–ligand binding affinity with applications to molecular docking. *Bioinformatics* **26**, 1169–1175 (2010).
27. Li, H., Leung, K.-S., Wong, M.-H. & Ballester, P. J. Substituting random forest for multiple linear regression improves binding affinity prediction of scoring functions: Cyscore as a case study. *BMC Bioinformatics* **15**, 291 (2014).
28. Li, Y., Zhang, X. & Cao, D. The role of shape complementarity in the protein–protein interactions. *Sci. Rep.* **3**, 3271 (2013).
29. Wallach, I., Dzamba, M. & Heifets, A. AtomNet: a deep convolutional neural network for bioactivity prediction in structure-based drug discovery. Preprint at <https://arxiv.org/abs/1510.02855> (2015).
30. Imrie, F., Bradley, A. R., van der Schaar, M. & Deane, C. M. Protein family-specific models using deep neural networks and transfer learning improve virtual screening and highlight the need for more data. *J. Chem. Inf. Model.* **58**, 2319–2330 (2018).
31. Balci, A. et al. DeepInterface: protein–protein interface validation using 3D convolutional neural networks. Preprint at *bioRxiv* <https://doi.org/10.1101/617506> (2019).
32. Wang, X., Terashi, G., Christoffer, C. W., Zhu, M. & Kihara, D. Protein docking model evaluation by 3D deep convolutional neural networks. *Bioinformatics* **36**, 2113–2118 (2020).
33. Renaud, N. et al. DeepRank: a deep learning framework for data mining 3D protein–protein interfaces. *Nat. Commun.* **12**, 7068 (2021).
34. Mohseni Behbahani, Y., Crouzet, S., Laine, E. & Carbone, A. Deep local analysis evaluates protein docking conformations with locally oriented cubes. *Bioinformatics* **38**, 4505–4512 (2022).
35. Kumawat, S. & Raman, S. LP-3DCNN: unveiling local phase in 3D convolutional neural networks. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition* 4903–4912 (IEEE, 2019).
36. Geng, C. et al. iScore: a novel graph kernel-based function for scoring protein–protein docking models. *Bioinformatics* **36**, 112–121 (2020).
37. Budowski-Tal, I., Kolodny, R. & Mandel-Gutfreund, Y. A novel geometry-based approach to infer protein interface similarity. *Sci. Rep.* **8**, 1–10 (2018).
38. Wang, X., Flannery, S. T. & Kihara, D. Protein docking model evaluation by graph neural networks. *Front. Mol. Biosci.* **8**, 647915 (2021).
39. Réau, M., Renaud, N., Xue, L. C. & Bonvin, A. M. DeepRank-GNN: a graph neural network framework to learn patterns in protein–protein interfaces. *Bioinformatics* **39**, 759 (2023).
40. Gainza, P. et al. Deciphering interaction fingerprints from protein molecular surfaces using geometric deep learning. *Nat. Methods* **17**, 184–192 (2020).
41. Sverrisson, F., Feydy, J., Correia, B. E. & Bronstein, M. M. Fast end-to-end learning on protein surfaces. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition* 15272–15281 (IEEE, 2021).
42. Zhai, X., Kolesnikov, A., Houlsby, N. & Beyer, L. Scaling vision transformers. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition* 12104–12113 (IEEE, 2022).
43. Dosovitskiy, A. et al. An image is worth 16x16 words: transformers for image recognition at scale. *9th International Conference on Learning Representations, ICLR* (2020).

44. Yan, Y., Tao, H., He, J. & Huang, S.-Y. The HDock server for integrated protein–protein docking. *Nat. Protoc.* **15**, 1829–1852 (2020).
45. Janin, J. et al. CAPRI: a critical assessment of predicted interactions. *Proteins* **52**, 2–9 (2003).
46. Khosla, P. et al. Supervised contrastive learning. *Ad. Neural Inf. Process. Syst.* **33**, 18661–18673 (2020).
47. Chen, C. et al. This looks like that: deep learning for interpretable image recognition. *Adv. Neural Inf. Process. Systems* **32**, 8930–8941 (2019).
48. Altschul, S. F. et al. Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. *Nucleic Acids Res.* **25**, 3389–3402 (1997).
49. Jones, D. T. Setting the standards for machine learning in biology. *Nat. Rev. Mol. Cell Biol.* **20**, 659–660 (2019).
50. Jeong, J.-J. et al. Characterization of the cupin-type phosphoglucose isomerase from the hyperthermophilic archaeon *Thermococcus litoralis*. *FEBS Lett.* **535**, 200–204 (2003).
51. Dominguez, C., Boelens, R. & Bonvin, A. M. HADDOCK: a protein–protein docking approach based on biochemical or biophysical information. *J. Am. Chem. Soc.* **125**, 1731–1737 (2003).
52. Wang, R., Fang, X., Lu, Y. & Wang, S. The PDBbind database: collection of binding affinities for protein–ligand complexes with known three-dimensional structures. *J. Med. Chem.* **47**, 2977–2980 (2004).
53. Evans, R. et al. Protein complex prediction with AlphaFold-Multimer. Preprint at *bioRxiv* <https://doi.org/10.1101/2021.10.04.463034> (2021).
54. Stebliankin, V. et al. EMOmIS: a pipeline for epitope-based molecular mimicry search in protein structures with applications to SARS-CoV-2. Preprint at *bioRxiv* <https://doi.org/10.1101/2022.02.05.479274> (2022).
55. Balbin, C. A. et al. Epitopedia: identifying molecular mimicry between pathogens and known immune epitopes. *Immunoinformatics* **9**, 100023 (2023).
56. Berman, H. M. et al. The Protein Data Bank. *Nucleic Acids Res.* **28**, 235–242 (2000).
57. Valdes, C. et al. Microbiome maps: Hilbert curve visualizations of metagenomic profiles. *Front. Bioinformatics* **3**, 1154588 (2023).
58. Andrusier, N., Nussinov, R. & Wolfson, H. J. FireDock: fast interaction refinement in molecular docking. *Proteins* **69**, 139–159 (2007).
59. Gray, J. J. et al. Protein–protein docking with simultaneous optimization of rigid-body displacement and side-chain conformations. *J. Mol. Biol.* **331**, 281–299 (2003).
60. Zhang, C., Vasmatzis, G., Cornette, J. L. & DeLisi, C. Determination of atomic desolvation energies from the structures of crystallized proteins. *J. Mol. Biol.* **267**, 707–726 (1997).
61. Dunbrack Jr, R. L. & Cohen, F. E. Bayesian statistical analysis of protein side-chain rotamer preferences. *Protein Sci.* **6**, 1661–1681 (1997).
62. Neria, E., Fischer, S. & Karplus, M. Simulation of activation free energies in molecular systems. *J. Chem. Phys.* **105**, 1902–1921 (1996).
63. Crowley, P. B. & Golovin, A. Cation– π interactions in protein–protein interfaces. *Proteins* **59**, 231–239 (2005).
64. Kabsch, W. & Sander, C. Dictionary of protein secondary structure: pattern recognition of hydrogen-bonded and geometrical features. *Biopolymers* **22**, 2577–2637 (1983).
65. Touw, W. G. et al. A series of PDB-related databanks for everyday needs. *Nucleic Acids Res.* **43**, 364–368 (2015).
66. Mead, A. Review of the development of multidimensional scaling methods. *J. R. Stat. Soc. Ser. D* **41**, 27–39 (1992).
67. Ng, A. Y. Feature selection, L_1 vs. L_2 regularization, and rotational invariance. In *Proc. Twenty-first International Conference on Machine Learning* 78 (Association for Computing Machinery, 2004).
68. Loshchilov, I. & Hutter, F. Decoupled weight decay regularization. *7th International Conference on Learning Representations, ICLR*, 6–9 (2017).
69. Lensink, M. F. & Wodak, S. J. Score_set: a CAPRI benchmark for scoring protein complexes. *Proteins* **82**, 3163–3169 (2014).
70. Zhang, C., Shine, M., Pyle, A. M. & Zhang, Y. US-align: universal structure alignments of proteins, nucleic acids, and macromolecular complexes. *Nat. Methods* **19**, 1109–1115 (2022).
71. Cheng, T. M.-K., Blundell, T. L. & Fernandez-Recio, J. pyDock: electrostatics and desolvation for effective scoring of rigid-body protein–protein docking. *Proteins* **68**, 503–515 (2007).
72. Leman, J. K. et al. Macromolecular modeling and design in Rosetta: recent methods and frameworks. *Nat. Methods* **17**, 665–680 (2020).
73. Pierce, B. & Weng, Z. A combination of rescoring and refinement significantly improves protein docking performance. *Proteins* **72**, 270–279 (2008).
74. Viswanath, S., Ravikant, D. & Elber, R. Improving ranking of models for protein complexes with side chain modeling and atomic potentials. *Proteins* **81**, 592–606 (2013).
75. Pons, C., Talavera, D., De La Cruz, X., Orozco, M. & Fernandez-Recio, J. Scoring by intermolecular pairwise propensities of exposed residues (SIPPER): a new efficient potential for protein–protein docking. *J. Chem. Inf. Model.* **51**, 370–377 (2011).
76. Ravikant, D. & Elber, R. Pie-efficient filters and coarse grained potentials for unbound protein–protein docking. *Proteins* **78**, 400–419 (2010).
77. Moal, I. H., Jiménez-García, B. & Fernández-Recio, J. CCharPPI web server: computational characterization of protein–protein interactions from structure. *Bioinformatics* **31**, 123–125 (2015).
78. Virtanen, P. et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nat. Methods* **17**, 261–272 (2020).
79. Renaud, N. & Geng, C. The pdb2sql Python package: parsing, manipulation and analysis of PDB files using SQL queries. *J. Open Source Softw.* **5**, 2077 (2020).
80. Paszke, A. et al. PyTorch: an imperative style, high-performance deep learning library. *Adv. Neural Inf. Processing Syst.* **32**, 8024–8035 (2019).
81. Collaborative Data Science (Plotly Technologies, 2015); <https://plot.ly>
82. Camacho, C. et al. BLAST+: architecture and applications. *BMC Bioinformatics* **10**, 421 (2009).
83. The PyMOL Molecular Graphics System, Version 1.8 (Schrödinger, LLC, 2015).
84. Stebliankin, V. stebliankin/piston: PlsToN (v1.0.0). *Zenodo* <https://doi.org/10.5281/zenodo.8102876> (2023).
85. Shore, D., Issafras, H., Landais, E., Teyton, L. & Wilson, I. The crystal structure of CD8 in complex with YTS156. 7.7 Fab and interaction with other CD8 antibodies define the binding mode of CD8 $\alpha\beta$ to MHC class I. *J. Mol. Biol.* **384**, 1190–1202 (2008).

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Author contributions

G.N. conceptualized and supervised the project. V.S. designed, implemented and tested the PlsToN framework. V.S. and A.S. executed benchmarks. P.B., J.S., P.C. and K.M. assisted in the feature extraction

design and interpretations. V.S., A.S., K.M. and G.N. contributed to the paper writing and rewriting.

Competing interests

The authors declare no competing interests.

Additional information

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